

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

End Semester Examination – Winter 2018

Course: B. Tech in Computer Science and Engg

Sem: III

Subject Name: Discrete Mathematics

Subject Code: BTCOC302

Date: 03/12/2018

Max Marks: 60

Duration: 3 Hrs.

Instructions to the Students:

1. Solve **ANY FIVE** questions out of the following.
2. Use of non-programmable scientific calculators is allowed.
3. Assume suitable data wherever necessary and mention it clearly.

Marks

Q. 1 Solve Any Three of the following.

- A) Let p and q be the propositions “Swimming at the New Jersey shore is allowed” and “Sharks have been spotted near the shore” respectively. Express each of these compound propositions as an English sentence. **4M**

- a) $\neg q$ b) $p \rightarrow \neg q$ c) $p \leftrightarrow \neg q$ d) $\neg p \vee q$

- B) Explain with example, notations used and mathematical expression to describe the following terms. **4 M**

- i) Membership ii) Subset iii) Equality of two sets iv) Union

- C) Use mathematical induction to show that $1+5+9+\dots+(4n-3) = n(2n-1)$, $\forall n \geq 1, n \in \mathbb{Z}$ **4M**

- D) Explain Universal quantifiers and Existential quantifiers with example. What is De Morgan's law for quantifiers? **4M**

Q.2 Solve the following.

- A) Check whether the relation R defined in the set $\{1, 2, 3, 4, 5, 6\}$ is **6 M**
 $R = \{(a, b) : b = a+1\}$ is reflexive, symmetric or transitive. Justify your answer. Find the relation Matrix.

- B) Explain surjective, injective, bijective and inverse function each with example. **6M**

Q. 3 Solve Any three of the following.

- A) Explain the pigeonhole principle with example. **4M**

- B) Find how many symbol codes can be formed if the first two symbols are letters and the next three are digits but no symbol is repeated? **4M**

- C) What is the expansion of $(3x + y)^4$? **4M**

- D) Determine the sequence $\{a_n\}$ where $a_n = 3n$ for every non-negative integer, n is a solution of the recurrence relation $a_n = 2a_{n-1} - a_{n-2}$ for $n = 2, 3, 4, \dots$ **4M**

Q.4 Solve the following.

- A) Define Euler graph and Hamiltonian Graph. **6M**

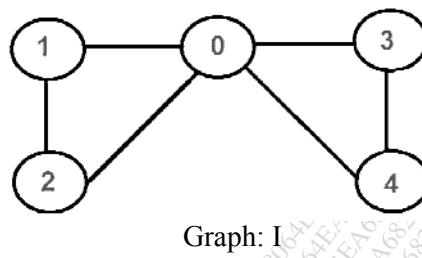
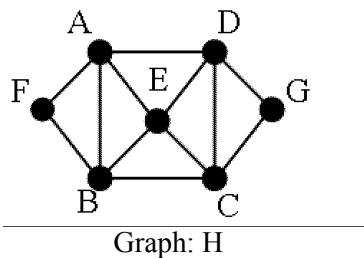
i) For a given graph G :

(a) Find a Hamiltonian path that begins at A and ends at E.

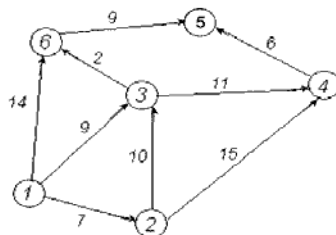
(b) Find a Hamiltonian circuit that starts at A and ends with the pair of vertices E, A.

(c) Find a Hamiltonian path that begins at F and ends at G.

ii) For a given graph I find Eulerian path and Eulerian cycle.



- B) Find the shortest path in the given graph using Dijkstra shortest path algorithm. **6M**

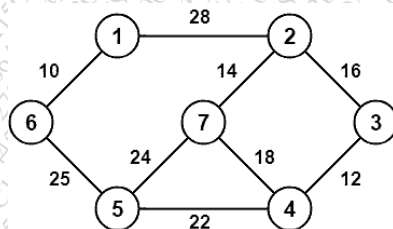


Q. 5 Solve Any three of the following.

- A) Show that a tree with n vertices has $n-1$ edges. **4M**
 B) Find minimum spanning tree for the given graph using Prim's algorithm? **4M**



- C) Define the following terms with reference to tree with example. **4M**
 i) Level and Height of a tree ii) M-ary Tree iii) Eccentricity of a vertex
 D) Construct the minimum spanning tree (MST) for the given graph using Kruskal's Algorithm. **4M**



Q. 6 Solve the following.

- A) Define the following terms. **6M**
 i) Algebraic Structures ii) Semi Groups iii) Monoids iv) Ring
 v) Field vi) Group
 B) For each of the following, determine whether the binary operation $*$ is commutative or associative? **6M**
 i) N is the set of natural numbers and $a * b = a + b + 2$ for $a, b \in N$
 ii) On N where $a * b = \min(a, b+2)$
 iii) On R where $a * b = a^b$

*** End ***